



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-41312

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Fri, Nov 29, 2024 at 12:20 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

29-Nov-2024

Dear Mr. Khare:

The manuscript entitled "Automatic Multi-point Turn-around Assistance System: Planning & Control" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access
ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Automatic Multi-point Turn-around Assistance System: Planning & Control

AUTHORS: Managheb, S. A. M.; Ghaffari, Ali

ABSTRACT: One of the daily challenges in automated driving is the turn-around task. This study describes the construction of an automatic turn-around assistance system(ATAS) that can perform single-or multi-point turns. This study highlights the hierarchical structure of the proposed system, which includes subsystems for control, path planning, and decision-making. The environment, including both narrow and wide roadways, a dual-axle vehicle model, and necessary sensors, is explained. Then, a low-cost decision-making subsystem based on reference paths is developed to complete this task with the least amount of steering efforts. Next, a novel analytical technique for path planning is put forward with a quick computation process. This geometry-based approach employs circular arcs and their conversion inspired by double-egg curves for any initial vehicle posture. Hence, continuous curvature paths are formed by hiring an alternative method rather than laborious Fresnel integral calculations. Finally, the control signals of longitudinal acceleration and steering angle are generated based on the arc length traveled throughout the mission. A safe, efficient, and comfy turn-around task is made possible by the ATAS, which satisfies the needs for versatility in various roads, smooth collision-free paths, and the less quantity of steering efforts. The MATLAB/Simulink and CarSim co-simulations assess the assistance system's capability on narrow and wide roads. With respectable precision, the CarSim model navigates both single and multi-point turns along the intended planned path. Furthermore, simulations illuminate that the proposed strategy outperforms conventional approaches like sampling and optimization-based methods in terms of mission execution time, computation time, and driving path quality attributes.

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